VINH THAI TONG

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EDUCATION

Bachelor of Science in Computer Science May 2027
The University of Texas at Dallas, Richardson, TX

- GPA 3.71/4.00
- Relevant Coursework: Data Structures and Algorithms, Database Systems, Advanced Algorithms, Artificial Intelligence, Machine Learning, Computer Networks.

EXPERIENCE

UT Dallas Office of Research and Innovation Richardson, TX
Research Information Systems Student Worker June 2026 - Present

- Collaborate on the development and maintenance of an interactive REDCap training resource using custom HTML, JavaScript, and CSS to improve navigation, content presentation, and usability.
- Manage, validate, and analyze 500+ research records across REDCap projects, troubleshoot technical and data-collection issues, and maintain technical documentation and knowledge base content.
- Develop and maintain a multi-page Power BI inventory dashboard connected to the REDCap API, using Power Query, DAX, and JSON to transform and visualize data while tracking device locations, movement history, and operational exceptions.

PROJECTS

PF Tracker | React, MongoDB, Auth0, Node.js, REST APIs, Gemini API, Recharts
HackUTD 2025: Lost in the Pages

- Developed a full-stack personal finance application for tracking income, expenses, net worth, transactions, cards, subscriptions, and category-based budgets.
- Built RESTful backend services with Node.js and Express.js, integrating MongoDB persistence, Auth0 JWT authentication, email verification, and environment-based configuration.
- Integrated a Gemini-powered financial assistant and interactive Recharts visualizations to provide personalized financial insights through a responsive interface.

Deep Learning Image Classification with MLPs and CNNs | Python, PyTorch, Torchvision, NumPy

- Developed a reusable PyTorch pipeline for classifying MNIST and CIFAR-10 images using three multilayer perceptron architectures and two convolutional neural networks.
- Conducted reproducible, validation-based hyperparameter tuning across learning rates, batch sizes, optimizers, dropout rates, and weight decay.
- Achieved 98.95% test accuracy on MNIST and demonstrated that CNNs substantially outperformed MLPs on the more complex CIFAR-10 dataset.

Enron Email Spam Classification | Python, NumPy, Logistic Regression, Naive Bayes, Bag-of-Words

- Built an end-to-end text-classification pipeline that transformed raw Enron emails into Bag-of-Words and Bernoulli feature representations.
- Implemented Logistic Regression from scratch using batch, mini-batch, and stochastic gradient descent with validation-based regularization tuning.
- Compared Logistic Regression, Multinomial Naive Bayes, and Bernoulli Naive Bayes using accuracy, precision, recall, and F1 score across multiple datasets.

LEADERSHIP

Vietnamese International Network of Culture, Education, and Friendship - VINCEF Richardson, TX
Cultural Chair June 2024 - June 2025

- Organized cultural programs engaging more than 50 students and received the Most Creative Award at UT Dallas's Passport to the World festival.
- Helped coordinate the organization's Global Talent Show performance, earning third place.

SKILLS

- **Languages:** Python, Java, C++, JavaScript, SQL, HTML, CSS, Assembly
- **Frameworks/Libraries:** React, Node.js, Express.js, PyTorch, Torchvision, NumPy, Recharts
- **Databases/APIs:** MongoDB, Mongoose, REST APIs, Gemini API, Auth0
- **Tools:** Git, Linux, Visual Studio Code, REDCap